ADVANCED PLACEMENT 40th Anniversary

AP Macroeconomics

FREE-RESPONSE

Scoring Guide with Multiple-Choice Section

1995

Advanced Placement Program®
THE COLLEGE BOARD

MACROECONOMICS

Two hours are allotted for this examination: 1 hour and 10 minutes for Section I, which consists of multiple-choice questions; and 50 minutes for Section II, which consists of three mandatory essay questions. Section I is printed in this examination booklet. Section II is printed in a separate booklet.

SECTION I

Time — 1 hour and 10 minutes

Number of questions — 60

Percent of total grade — 66 2/3

Section I of this examination contains 60 multiple-choice questions. Therefore, please be careful to fill in only the ovals that are preceded by numbers 1 to 60 on your answer sheet.

General Instructions

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE INSTRUCTED TO DO SO.

INDICATE ALL YOUR ANSWERS TO QUESTIONS IN SECTION I ON THE SEPARATE ANSWER SHEET. No credit will be given for anything written in this examination booklet, but you may use the booklet for notes or scratchwork. After you have decided which of the suggested answers is best, COMPLETELY fill in the corresponding oval on the answer sheet. Give only one answer to each question. If you change an answer, be sure that the previous mark is erased completely.

Example:

Chicago is a
(A) state
(B) city
(C) country
(D) continent
(E) village

Sample Answer

Ⓐ ● Ⓒ Ⓓ Ⓔ

Many candidates wonder whether or not to guess the answers to questions about which they are not certain. In this section of the examination, as a correction for haphazard guessing, one-fourth of the number of questions you answer incorrectly will be subtracted from the number of questions you answer correctly. It is improbable, therefore, that mere guessing will improve your score significantly; it may even lower your score, and it does take time. If, however, you are not sure of the correct answer but have some knowledge of the question and are able to eliminate one or more of the answer choices as wrong, your chance of getting the right answer is improved, and it may be to your advantage to answer such a question.

Use your time effectively, working as rapidly as you can without losing accuracy. Do not spend too much time on questions that are too difficult. Go on to other questions and come back to the difficult ones later if you have time. It is not expected that everyone will be able to answer all the multiple-choice questions.

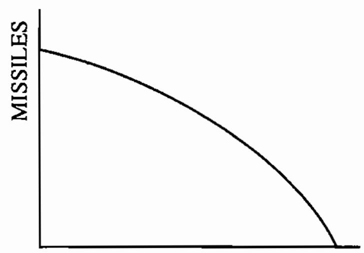
MACROECONOMICS

SECTION I

Time — 70 minutes

60 Questions

Directions: Each of the questions or incomplete statements below is followed by five suggested answers or completions. Select the one that is best in each case and then fill in the corresponding oval on the answer sheet.



1. Which of the following would cause the production possibilities curve shown above to shift outward?
   (A) Reopening steel plants that had been closed
   (B) Rehiring laid-off steelworkers
   (C) Using machinery for missile production instead of steel production
   (D) Using machinery for steel production instead of missile production
   (E) Developing a more efficient steelmaking process

2. If in a specified year nominal gross domestic product grew by 11 percent and real gross domestic product grew by 4 percent, inflation for this year would be
   (A) −7%
   (B) 7%
   (C) 8%
   (D) 11%
   (E) 15%

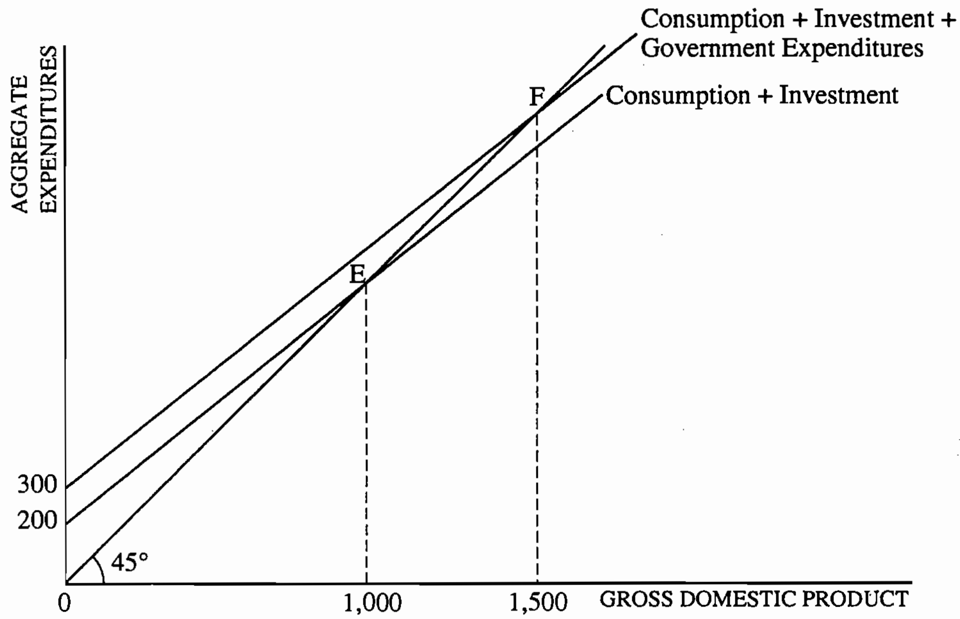
3. Which of the following is an example of structural unemployment?
   (A) A computer programmer who leaves her job to move to Florida
   (B) A worker who loses his job during a recession
   (C) An autoworker who is replaced by a robot
   (D) A construction worker who is always unemployed during the winter months
   (E) A worker who is engaged in unproductive work

4. What would be the effect of a large increase in labor productivity on the real gross domestic product and the price level?

	Real Gross Domestic Product	Price Level
(A)	Increase	Increase
(B)	Increase	Decrease
(C)	No effect	Increase
(D)	Decrease	Increase
(E)	Decrease	Decrease

5. An increase in which of the following will increase the value of the spending multiplier?
   (A) The supply of money
   (B) Equilibrium output
   (C) Personal income tax rates
   (D) The marginal propensity to consume
   (E) The required reserve ratio

6. According to the Keynesian model, which of the following would increase aggregate demand?
   (A) An increase in autonomous investment
   (B) An increase in the discount rate
   (C) A decrease in unemployment compensation payments
   (D) A decrease in government expenditures accompanied by an equal reduction in taxes
   (E) A decrease in government expenditures on public works



7. The graph above indicates equilibrium at E for a closed economy without government spending. If the addition of government spending results in equilibrium at F, which of the following is true?

(A) Government spending is \$300 and the multiplier is 5.
(B) Government spending is \$100 and the multiplier is 5.
(C) Government spending is \$100 and consumption increases by \$500.
(D) Government spending and gross domestic product increase by \$500 each.
(E) Consumption and gross domestic product increase by \$500 each.

8. Commercial banks can create money by

   (A) transferring depositors' accounts at the Federal Reserve for conversion to cash
   (B) buying Treasury bills from the Federal Reserve
   (C) sending vault cash to the Federal Reserve
   (D) maintaining a 100 percent reserve requirement
   (E) lending excess reserves to customers

9. If the reserve requirement is 20 percent, the existence of $100 worth of excess reserves in the banking system can lead to a maximum expansion of the money supply equal to

   (A) $20
   (B) $100
   (C) $300
   (D) $500
   (E) $750

10. If the Federal Reserve lowers the reserve requirement, which of the following would most likely occur?

    (A) Imports will rise, decreasing the trade deficit.
    (B) The rate of saving will increase.
    (C) Unemployment and inflation will both increase.
    (D) Businesses will purchase more factories and equipment.
    (E) The budget deficit will increase.

11. If the public's desire to hold money as currency increases, what will the impact be on the banking system?

    (A) Banks would be more able to reduce unemployment.
    (B) Banks would be more able to decrease aggregate supply.
    (C) Banks would be less able to decrease aggregate supply.
    (D) Banks would be more able to expand credit.
    (E) Banks would be less able to expand credit.

12. According to Keynesian theory, decreasing taxes and increasing government spending will most likely change consumption expenditures and unemployment in which of the following ways?

	Consumption Expenditures	Unemployment
(A)	Decrease	Increase
(B)	Decrease	No change
(C)	Increase	Decrease
(D)	Increase	Increase
(E)	No change	Decrease

13. Which of the following policy combinations is most likely to cure a severe recession?

	Open-Market Operations	Taxes	Government Spending
(A)	Buy securities	Increase	Decrease
(B)	Buy securities	Decrease	Increase
(C)	Buy securities	Decrease	Decrease
(D)	Sell securities	Decrease	Decrease
(E)	Sell securities	Increase	Increase

14. In an economy at full employment, a presidential candidate proposes cutting the government debt in half in four years by increasing income tax rates and reducing government expenditures. According to Keynesian theory, implementation of these policies is most likely to increase

(A) unemployment
(B) consumer prices
(C) aggregate demand
(D) aggregate supply
(E) the rate of economic growth

15. To protect high-cost domestic producers, a country imposes a tariff on an imported commodity, Y. Which of the following is most likely to occur in the short run?

I. A decrease in domestic production of Y
II. An increase in domestic production of Y
III. An increase in foreign output of Y

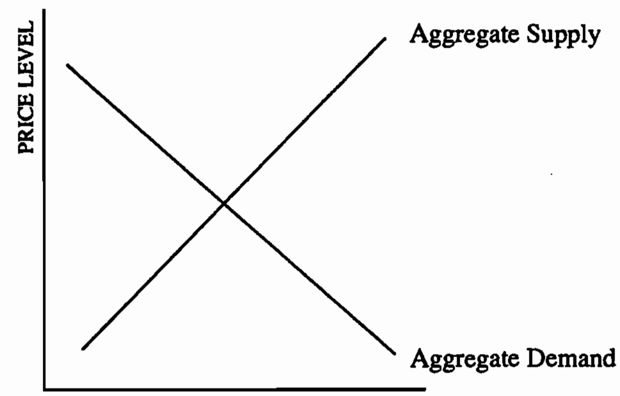
(A) I only
(B) II only
(C) III only
(D) I and III only
(E) II and III only

16. Suppose two countries are each capable of individually producing two given commodities. Instead, each specializes by producing the commodity for which it has a comparative advantage and then trades with the other country. Which of the following is most likely to result?

(A) The two countries will become more independent of each other.
(B) Unemployment will increase in one country and decrease in the other.
(C) There will be more efficient production in one country but less efficient production in the other.
(D) Both countries will become better off.
(E) Both countries will be producing their commodity inefficiently.

17. Which of the following means of reducing military spending would have the greatest positive impact on gross domestic product for the United States?

(A) Combining two domestic military bases into one overseas base
(B) Cutting retirement benefits to military personnel
(C) Closing overseas military bases and relocating those operations to the United States
(D) Closing overseas military bases and laying off military personnel
(E) Canceling contracts with domestic producers for new airplanes



18. According to the graph above, which of the following will necessarily result in a decrease in output?

I. A rightward shift of the aggregate demand curve
II. A leftward shift of the aggregate demand curve
III. A rightward shift of the aggregate supply curve
IV. A leftward shift of the aggregate supply curve

(A) I only
(B) III only
(C) I and III only
(D) II and III only
(E) II and IV only

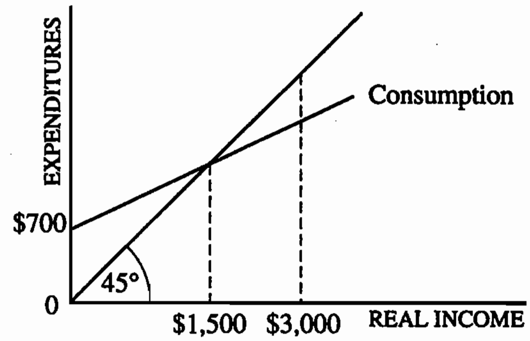
19. Which of the following will result in the greatest increase in aggregate demand?

   (A) A $100 increase in taxes
   (B) A $100 decrease in taxes
   (C) A $100 increase in government expenditures
   (D) A $100 increase in government expenditures, coupled with a $100 increase in taxes
   (E) A $100 increase in government expenditures, coupled with a $100 decrease in taxes

20. If the economy is in a severe recession, which of the following is the fiscal policy most effective in stimulating production?

   (A) Government spending increases.
   (B) Government spending decreases.
   (C) Personal income taxes are increased.
   (D) The Federal Reserve sells bonds on the open market.
   (E) The Federal Reserve buys bonds on the open market.

<u>Questions 21-22</u> refer to the diagram below, which depicts an economy's consumption function.



21. If the marginal propensity to consume increases, the equilibrium levels of income and consumption will change in which of the following ways?

	Equilibrium Level of Income	Equilibrium Level of Consumption
(A)	No change	No change
(B)	No change	Increase
(C)	Increase	No change
(D)	Increase	Increase
(E)	Decrease	Decrease

22. If private investment of $100 is added to the economy, the equilibrium levels of income and consumption will change in which of the following ways?

	Equilibrium Level of Income	Equilibrium Level of Consumption
(A)	Increase	Decrease
(B)	Increase	Increase
(C)	Increase	No change
(D)	No change	Increase
(E)	No change	No change

23. The real value of the United States dollar is determined by

(A) federal regulations regarding purchasing power
(B) the value of the gold backing the dollar
(C) the goods and services it will buy
(D) the money multiplier
(E) the marginal propensity to consume

24. The demand for money increases when national income increases because

(A) spending on goods and services increases
(B) interest rates increase
(C) the budget deficit increases
(D) the money supply increases
(E) the public becomes more optimistic about the future

25. An aggregate supply curve may be horizontal over some range because within that range

(A) a higher price level leads to higher interest rates, which reduce the money supply and consumer spending
(B) changes in the aggregate price level do not induce substitution
(C) output cannot be increased unless prices and interest rates increase
(D) rigid prices prevent employment from fluctuating
(E) resources are underemployed and an increase in demand will be satisfied without any pressure on the price level

26. If the Federal Reserve wishes to use monetary policy to reinforce Congress' fiscal policy changes, it should

(A) increase the money supply when government spending is increased
(B) increase the money supply when government spending is decreased
(C) decrease the money supply when government spending is increased
(D) increase interest rates when government spending is increased
(E) decrease interest rates when government spending is decreased

27. Which of the following relationships is illustrated by a short-run Phillips curve?

(A) A decrease in the rate of inflation is accompanied by an increase in the rate of economic growth.
(B) A decrease in the rate of inflation is accompanied by an increase in the rate of unemployment.
(C) An increase in the rate of inflation is accompanied by a decrease in the rate of economic growth.
(D) An increase in the rate of inflation is accompanied by an increase in the rate of unemployment.
(E) A decrease in the rate of economic growth is accompanied by a decrease in the rate of unemployment.

28. Which of the following could cause simultaneous increases in inflation and unemployment?

(A) A decrease in government spending
(B) A decrease in the money supply
(C) A decrease in the velocity of money
(D) An increase in inflationary expectations
(E) An increase in the overall level of productivity

29. If other things are held constant, an increase in United States imports will

(A) tend to cause the dollar to appreciate because the world supply of dollars will rise
(B) tend to cause the dollar to appreciate because the world demand for dollars will rise
(C) have no effect on the exchange rate for the dollar because exports will also increase
(D) tend to cause the dollar to depreciate because the world supply of dollars will rise
(E) tend to cause the dollar to depreciate because the world demand for dollars will rise

30. An increase in which of the following is most likely to cause an improvement in the standard of living over time?

(A) Size of the population
(B) Size of the labor force
(C) Number of banks
(D) Level of taxation
(E) Productivity of labor

31. The long-run aggregate supply curve is likely to shift to the right when there is

(A) an increase in the cost of productive resources
(B) an increase in productivity
(C) an increase in the federal budget deficit
(D) a decrease in the money supply
(E) a decrease in the labor force

32. The consumer price index measures which of the following?

(A) The change over time of the weighted prices of a particular group of goods and services
(B) The change over time of the weighted wholesale price index
(C) The change over time of the difference between the gross domestic product deflator and the wholesale price index
(D) Inflation corrected for changes in the real gross domestic product
(E) Inflation corrected for changes in the wholesale price index

33. Which of the following is true if the economy is producing at the full-employment level of output?

(A) The unemployment rate is zero.
(B) No person is receiving unemployment compensation from the government.
(C) There is frictional unemployment.
(D) The government's budget is balanced.
(E) The balance of trade is in equilibrium.

34. Which of the following is a basic tenet of classical economic analysis?

(A) Saving is usually greater than investment.
(B) The economy is self-correcting to full employment.
(C) The economy may be in equilibrium at less than full employment.
(D) Inflation is not a serious economic problem.
(E) The prices of products tend to be inflexible.

35. Which of the following will most likely result from a decrease in government spending?

(A) An increase in output
(B) An increase in the price level
(C) An increase in employment
(D) A decrease in aggregate supply
(E) A decrease in aggregate demand

36. Current equilibrium output equals \$2,500,000, potential output equals \$2,600,000, and the marginal propensity to consume equals 0.75. Under these conditions, a Keynesian economist is most likely to recommend

(A) decreasing taxes by \$25,000
(B) decreasing taxes by \$100,000
(C) increasing government spending by \$25,000
(D) increasing government spending by \$33,333
(E) increasing government spending by \$100,000

37. An inflationary gap could be reduced by

(A) an increase in government spending
(B) an increase in the supply of money
(C) an increase in the income tax rate
(D) a decrease in the discount rate
(E) a decrease in the reserve requirement

38. The circular flow of economic activity between consumers and producers includes which of the following?

I. Households buy factor services from firms.
II. Households sell factor services to firms.
III. Households buy outputs from firms.
IV. Households sell outputs to firms.

(A) III only
(B) IV only
(C) I and II only
(D) II and III only
(E) III and IV only

39. Suppose the required reserve ratio is 20 percent and a single bank with no excess reserves receives a $100 deposit from a new customer. The bank now has excess reserves equal to

(A) $20
(B) $80
(C) $100
(D) $400
(E) $500

40. Which of the following is most likely to increase if the public decides to increase its holdings of currency?

(A) The interest rate
(B) The price level
(C) Disposable personal income
(D) Employment
(E) The reserve requirement

41. During a mild recession, if policymakers want to reduce unemployment by increasing investment, which of the following policies would be most appropriate?

(A) Equal increases in government expenditure and taxes
(B) An increase in government expenditure only
(C) An increase in transfer payments
(D) An increase in the reserve requirement
(E) Purchase of government securities by the Federal Reserve

42. Which of the following monetary and fiscal policy combinations would most likely result in a decrease in aggregate demand?

	Discount Rate	Open-Market Operations	Government Spending
(A)	Lower	Buy bonds	Increase
(B)	Lower	Buy bonds	Decrease
(C)	Raise	Sell bonds	Increase
(D)	Raise	Buy bonds	Increase
(E)	Raise	Sell bonds	Decrease

43. Which of the following is true of supply shocks?

(A) They tend to change both relative prices and the general price level in the economy.
(B) They affect only the general price level.
(C) They can be anticipated and offset with appropriate fiscal policy.
(D) They can be anticipated and offset with appropriate monetary policy.
(E) They make the aggregate supply curve vertical.

44. Suppose that, from 1985 to 1986, unemployment fell from 7.2 to 7.0 percent and inflation fell from 3.8 to 1.1 percent. An explanation of these changes might be that the

(A) aggregate demand curve shifted to the left
(B) aggregate demand curve shifted to the right
(C) aggregate supply curve shifted to the left
(D) aggregate supply curve shifted to the right
(E) short-run Phillips curve shifted to the right

45. If higher United States interest rates cause foreign demand for the dollar to increase, which of the following will occur to the international value of the dollar and to United States exports?

	International Value of the Dollar	Exports
(A)	Increase	Increase
(B)	Increase	Decrease
(C)	Increase	No change
(D)	Decrease	Increase
(E)	Decrease	Decrease

46. The table below indicates the number of labor hours required in Countries X and Y to produce one unit of food or one unit of clothing.

Country	Food	Clothing
X	20 hours	50 hours
Y	10 hours	20 hours

Given this information, which of the following statements is correct?

(A) X has a comparative advantage in the production of both food and clothing.
(B) Y has a comparative advantage in the production of both food and clothing.
(C) X has a comparative advantage in food production, whereas Y has a comparative advantage in clothing production.
(D) Y has a comparative advantage in food production, whereas X has a comparative advantage in clothing production.
(E) Neither country has a comparative advantage in the production of either good.

47. Which of the following groups of people would benefit from unanticipated inflation?

I. Savers
II. Borrowers
III. Lenders

(A) I only
(B) II only
(C) III only
(D) I and II only
(E) I and III only

48. An increase in the labor force participation rate will

(A) increase investment and decrease savings
(B) increase savings and decrease investment
(C) have no effect on unemployment
(D) make it easier to reduce unemployment
(E) make it more difficult to reduce unemployment

49. Which of the following is a key feature of Keynesian economics?

(A) The level of saving depends mostly on interest rates.
(B) The level of government expenditure depends mostly on interest rates.
(C) Supply creates its own demand.
(D) Macroeconomic equilibrium can occur at less than full employment.
(E) Wages are more flexible than prices.

50. If a large increase in total spending has no effect on real gross domestic product, it must be true that

(A) the price level is rising
(B) the economy is experiencing high unemployment
(C) the spending multiplier is equal to 1
(D) the economy is in short-run equilibrium
(E) aggregate supply has increased

51. According to Keynesian theory, the most important determinant of saving and consumption is the

(A) interest rate
(B) price level
(C) level of income
(D) level of employment
(E) flexibility of wages and prices

52. Under which of the following circumstances would increasing the money supply be most effective in increasing real gross domestic product?

	Interest Rates	Employment	Business Optimism
(A)	High	Full	High
(B)	High	Less than full	High
(C)	Low	Full	High
(D)	Low	Full	Low
(E)	Low	Less than full	Low

53. Faced with a large federal budget deficit, the government decides to decrease expenditures and tax revenues by the same amount. This action will affect output and interest rates in which of the following ways?

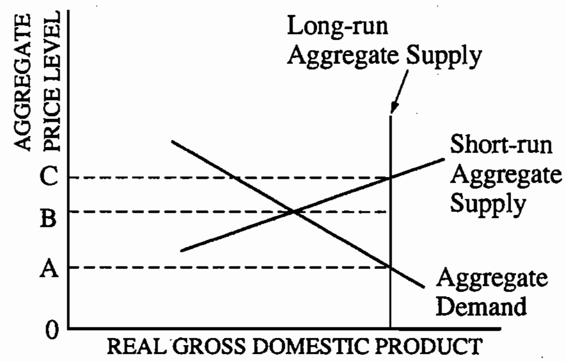
	Output	Interest Rates
(A)	Increase	Increase
(B)	Increase	Decrease
(C)	No change	Decrease
(D)	Decrease	Increase
(E)	Decrease	Decrease

54. If crowding out only partially offsets the effects of a tax cut, which of the following changes in interest rates and gross domestic product are most likely to occur?

	Interest Rates	Gross Domestic Product
(A)	Increase	Increase
(B)	Increase	Remain unchanged
(C)	Increase	Decrease
(D)	Remain unchanged	Increase
(E)	Decrease	Decrease

55. All of the following are components of the money supply in the United States EXCEPT

(A) paper money
(B) gold bullion
(C) checkable deposits
(D) coins
(E) demand deposits



56. The graph above depicts an economy's aggregate demand and aggregate supply curves. If aggregate demand remains constant, the equilibrium price levels in the short run and in the long run will be which of the following?

	Short Run	Long Run
(A)	0A	0A
(B)	0B	0A
(C)	0B	0C
(D)	0C	0A
(E)	0C	0C

57. According to both monetarists and Keynesians, which of the following happens when the Federal Reserve reduces the discount rate?

(A) The demand for money decreases and market interest rates decrease.
(B) The demand for money increases and market interest rates increase.
(C) The supply of money increases and market interest rates decrease.
(D) The supply of money increases and market interest rates increase.
(E) Both the demand for money and the supply of money increase and market interest rates increase.

58. An increase in which of the following is most likely to increase the long-run growth rate of an economy's real per capita income?

(A) Population growth
(B) The proportion of gross domestic product consumed
(C) The educational attainment of the population
(D) The supply of money in circulation
(E) Personal income taxes

59. An increase in the money supply will have the greatest effect on real gross domestic product if

(A) the marginal propensity to consume is low
(B) unemployment is very low
(C) investment spending is not sensitive to changes in interest rates
(D) the quantity of money demanded is not very sensitive to interest rates
(E) the required reserve ratio is high

60. If the Federal Reserve undertakes a policy to reduce interest rates, international capital flows will be affected in which of the following ways?

(A) Long-run capital outflows from the United States will decrease.
(B) Long-run capital inflows to the United States will increase.
(C) Short-run capital outflows from the United States will decrease.
(D) Short-run capital inflows to the United States will decrease.
(E) Short-run capital inflows to the United States will not change.

MACROECONOMICS

SECTION II

Time — 50 minutes

Directions: You have fifty minutes to answer all three of the following questions. It is suggested that you take a few minutes to plan and outline each answer. Spend approximately half your time on the first question and divide the remaining time equally between the next two questions. In answering the questions, you should emphasize the line of reasoning that generated your results; it is not enough to list the results of your analysis. Include diagrams, if useful, in explaining your answers. All diagrams should be clearly labeled.

1. Over the past two years, the unemployment rate in Country X has risen from 5 percent to 9 percent. As the leader of Country X, you have been presented with two policy options to address the unemployment problem.

   Policy 1: Use tariffs and quotas to restrict imports and thus protect jobs in Country X.
   Policy 2: Use monetary and fiscal policies to solve the unemployment problem without resorting to trade restrictions.

   (a) Explain two disadvantages of selecting Policy 1.
   (b) Describe in detail one specific monetary policy action and one specific fiscal policy action you would take to reduce unemployment. Explain how each of these actions would affect each of the following in the short run.
      (i) Aggregate demand
      (ii) Output and the price level
      (iii) Real interest rates
   (c) If the interest rate effects you identified in Part (b) continue in the long run, explain the impact of these effects on economic growth.

2. Assume that the economy is in a recession.
   (a) Explain each of the following.
      (i) Monetary and fiscal policies advocated by monetarists to eliminate the recession
      (ii) Monetary and fiscal policies advocated by Keynesians to eliminate the recession
   (b) Explain how monetarists and Keynesians differ in their conclusions about the effects of crowding out associated with the stabilization policies outlined in Part (a).

3. Explain how some individuals are helped and others harmed by unanticipated inflation as they participate in each of the following markets.
   (a) Credit markets
   (b) Labor markets
   (c) Product markets

Chapter III

Answers to the 1995 AP Macroeconomics Examination

■ SECTION I: MULTIPLE-CHOICE

Listed below are the correct answers to the multiple-choice questions and the percentage of AP candidates who attempted each question and answered it correctly. An answer sheet gridded with the correct responses appears on the next page.

Section I Answer Key and Percent Answering Correctly

Item No.	Correct Answer	Percent Correct	Item No.	Correct Answer	Percent Correct
1	E	83%	31	B	88%
2	B	87%	32	A	58%
3	C	71%	33	C	70%
4	B	73%	34	B	65%
5	D	71%	35	E	65%
6	A	54%	36	C	48%
7	B	61%	37	C	70%
8	E	82%	38	D	75%
9	D	65%	39	B	76%
10	D	71%	40	A	45%
11	E	61%	41	E	47%
12	C	84%	42	E	73%
13	B	86%	43	A	46%
14	A	79%	44	D	45%
15	B	76%	45	B	57%
16	D	90%	46	C	26%
17	C	79%	47	B	60%
18	E	81%	48	E	35%
19	E	85%	49	D	46%
20	A	79%	50	A	59%
21	D	56%	51	C	35%
22	B	48%	52	B	35%
23	C	76%	53	E	27%
24	A	61%	54	A	28%
25	E	61%	55	B	79%
26	A	73%	56	B	67%
27	B	59%	57	C	57%
28	D	45%	58	C	52%
29	D	65%	59	D	19%
30	E	85%	60	D	47%

Free-Response Question 1

Over the past two years, the unemployment rate in Country X has risen from 5 percent to 9 percent. As the leader of Country X, you have been presented with two policy options to address the unemployment problem.

Policy 1: Use tariffs and quotas to restrict imports and thus protect jobs in Country X.
Policy 2: Use monetary and fiscal policies to solve the unemployment problem without resorting to trade restrictions.

(a) Explain two disadvantages of selecting Policy 1.
(b) Describe in detail one specific monetary policy action and one specific fiscal policy action you would take to reduce unemployment. Explain how each of these actions would affect each of the following in the short run.
  (i) Aggregate demand
  (ii) Output and the price level
  (iii) Real interest rates
(c) If the interest rate effects you identified in Part (b) continue in the long run, explain the impact of these effects on economic growth.

Question 1 Scoring Guide

Basically the point distribution is 2 points for Part (a); 5 for Part (b); and 2 for Part (c).

Part (a): 2 points

Two disadvantages to tariffs and quotas may include:

- domestic prices will rise resulting in a lower standard of living;
- trade war/retaliation resulting in increased domestic prices and unemployment;
- if foreign income decreases due to lost domestic market of Country X, resulting in a decreased demand for exports of Country X, then unemployment in Country X will increase;
- increase in unemployment in import dependent industries;
- loss of economic efficiency (comparative advantage explanation);
- reduced competition in Country X implying increased market power in domestic industries;

Points:
1 point per disadvantage and explanation.

Part (b): 5 points

Expansionary monetary *and* fiscal policies are required. Expansionary monetary policy must include one of the following: OMO purchases, reduction in reserve requirement, or reduction in discount rate. Monetary policy leads to an increase in the money supply; a decrease in real interest rates; an increase in investment which is a component of Aggregate Demand; AD increases, resulting in increased output and prices. Fiscal policies are increasing government spending or decreasing taxes (corporate or personal income). An increase in G will directly increase AD resulting in an increase in output and prices. A decrease in taxes will increase C (or I if corporate taxes discussed) and directly increase AD. Output, employment, and prices will rise. Real interest rates will rise because the increase in government spending increases the demand for funds in the bond market (or loanable funds market). An alternative explanation is that nominal interest rates will rise because the expansionary fiscal policy increases income which, in turn, increases the demand for money. The effect on real interest rates is ambiguous because the magnitude of the price increase relative to the nominal interest rates' increase is unknown.

Points:
1 — Correct stabilization policies (a ½ point for each policy).
2 — Monetary policy explanation of linkages of real interest rate-investment-AD, output and price.

$$1 \quad \begin{cases} \uparrow M^s \rightarrow \downarrow i \\ \downarrow i \rightarrow \uparrow I \end{cases}$$

$$1 \quad \begin{cases} \uparrow I \rightarrow \uparrow AD \\ \uparrow AD \rightarrow \uparrow Q, \uparrow P \end{cases}$$

1 — Fiscal policy linkages of G/TX resulting in AD, output and price explanation.
1 — An explanation of the fiscal policy effect on real interest rates.

Part (c): 2 points

There are contradictory effects of interest rates as a result of the two stabilization policies — this should be noted by the student; however, consistency with answer to Part (b) is paramount. Decreasing interest rates will encourage investment which will stimulate capital stock growth or economic growth which in turn will shift the long-run aggregate supply curve or the production possibilities curve/frontier. [Positive net investment increases or investment in excess of replacement investment increases.] Increasing interest rates will have just the reverse effect — including a leftward shift in the long run aggregate supply curve or production possibilities curve.

Points:

1 — For the linkage of interest => investment => economic growth. Interest rate direction must be consistent with answer in Part (b).

1 — Shift in long run aggregate supply curve or the production possibilities curve.

Alternative Acceptable Answers:

- The monetarists would say that fiscal policy would not be as effective as Keynesians would and that it was probably fiscal policy which got us into the current situation. The monetarists would use the monetary policy tools to maintain a constant money supply growth to ensure a stable financial environment to permit the economy to self-adjust toward full employment GDP.
- Supply-siders would suggest that a decrease in taxes would provide incentives for increasing productivity and thus the AS curve would shift rightward. An accompanying AD shift is required for full credit.

To receive credit for an alternative answer, the student had to specify that the answer was using a particular theory. Other economic schools of thought were accepted and evaluated; however, no other school of thought generated a sufficient number of answers to provide a rubric. These answers were evaluated by the table leader.

Besides counting points, the answer may be looked at as a whole and ultimately judged by its overall quality. This is particularly true if the total point count includes a half. The final total should mean something in terms of the overall quality of the answer: 8 or 9 should reflect an excellent answer (a 9 is not necessarily a perfect answer); 6 and 7, a good answer; 4 and 5, an adequate answer; 3, a seriously deficient answer, but still an answer; 2, answers that are lots of words signifying nothing except one sustained argument; 1, a correct, relevant-to-the-question statement; 0, no relevant economic answer to the question. Using the holistic approach, the 1 or 2 is a "bottom up" approach.

Overall Comment on Student Responses to Question 1

Students continued to experience difficulty in addressing the impact of fiscal policy on interest rates and economic growth questions.

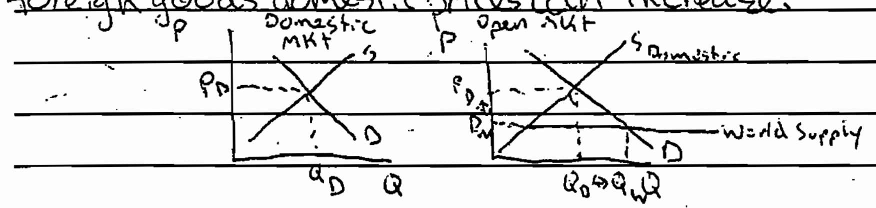
The primary mechanism by which fiscal policy affects interest rates is through the loanable funds market. If the government desires to increase government spending or decrease taxes, the resulting budget deficit must be funded through the sale of government securities. When the Treasury enters the loanable funds market, the demand for loanable funds increases and the interest rate rises. The reverse occurs with contractionary fiscal policy. Many students address the interest rate effect of fiscal policy through the money demand function. The standard line of reasoning is that increases in government spending increase aggregate demand which in turn increase equilibrium income. The increase in income will increase the demand for money thereby increasing interest rates. The above reasoning is correct; however, it is a *secondary* effect, which applies to expansionary monetary policy as well as to expansionary fiscal policy.

On the issue of economic growth, students confuse increases in GDP brought about by increases in aggregate demand with economic growth. Economic growth involves the change in *productive capacity*. We can represent economic growth by a change in potential GDP, changes in the long-run aggregate supply curve, or shifts in the production possibilities curve. Merely stating that an increase in investment leads to economic growth is not sufficient. The increase in investment must be beyond replacement investment; alternatively, the student could state the effect as an increase in net investment.

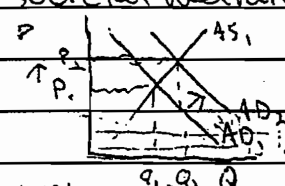
Sample Student Responses

Student Response 1 (Score of 9)

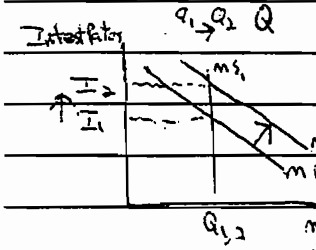
a) Using quotas and tariffs to restrict trade would only, at best, produce a short-run rise in employment. Temporarily the problem may be transferred to another country (beggar-thy-neighbor), but in the long run the standard of living in country X would actually decrease. By specializing and pursuing comparative advantages, a country is allowed to operate outside ~~them~~ its production possibilities curve. The input of foreign goods actually increases employment because large numbers of people would be employed by efficient exporting industries. Most likely, the output of these industries would decrease when other countries retaliated to the restrictions. This policy would also cause a rise in domestic price level. Without the competition of foreign goods domestic prices can increase.



b) Increasing government spending would be a powerful way to decrease unemployment. The government spending multiplier, $\frac{1}{1-\text{Marg. prop. to consum}}$ allows a small increase in spending to translate to a large increase in societal welfare.



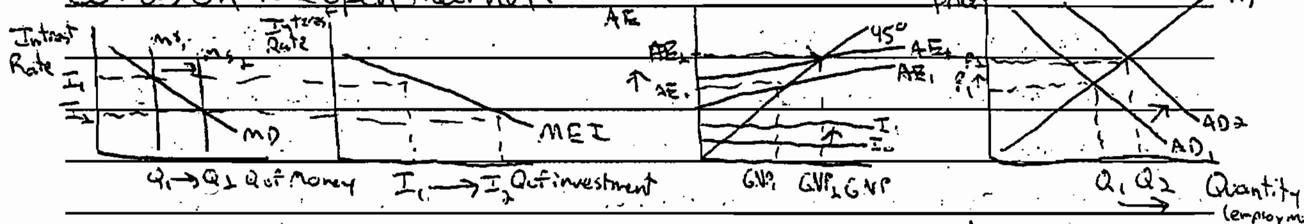
The increase in spending would cause an increase in Aggregate demand which causes an increase in both price levels and output.



As shown in the diagram, the nominal interest rates would increase because of an increase in Money demand, but this increase would be somewhat offset by the increase in inflation (Nom. Interest Rate = Real Interest + Inflation (expected))

So in reality, real interest rates would probably not be affected greatly.

The most commonly used monetary policy option is Open Market Operations. To reduce unemployment, there needs to be an increase in the money supply. This would occur if the fed bought government bonds on the open market.



This fiscal policy forces lower interest rates which translate to an increase in investment. This increases aggregate expenditure which is aggregate demand, that is aggregate demand rises. This entails an increase in both output and price level. Nominal interest rates fall and inflation rises. This translates to a drop in the real interest rate because Real interest rate = Nominal - Expected inflation.

C) The low interest rates stimulate investment. This investment, if for instance was in technology, would increase a society's production possibilities frontier. In the long run this heightened investment would cause an increase in long run growth.

Comment: The essay receives full credit in Part (a) with the arguments of a decreased standard of living and the effects of retaliatory tariffs and quotas. The discussion of fiscal policy meets and exceeds the standards by supplying the information concerning the government spending multiplier. The interest rate effect of fiscal policy is via the money demand function, which in reality is a secondary effect, which also exists in expansionary monetary policy. The preferred interest rate effect is that of increased demand for loanable funds which occurs with the government financing the increased expenditures through the sale of Treasury securities. The expansionary monetary policy discussion is complete, although the student mistakenly refers to increases in the money supply via open market operations as fiscal policy. In Part (c), the student relates the low interest rates to stimulating investment which increases technology to shift the production possibilities curve outward, thus demonstrating an understanding of economic growth. This answer has two characteristics which are important to underscore: first, the student uses graphical analysis to enhance the answers and to ensure that the reader understands the verbal presentation; second, while the student did misstate fiscal policy instead of monetary policy, the student still received full credit. (A 9 is not necessarily a perfect answer.)

Student Response 2 (Score of 8)

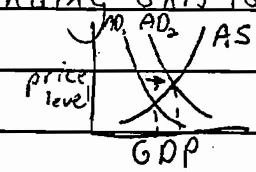
a) Policy I has 2 disadvantages:

1.) trade between countries is reduced because of quotas therefore comparative advantage in each country is not occuring (ie I country is forced to produce a product that the other country could produce more efficiently) therefore efficiency can decline in both countries.

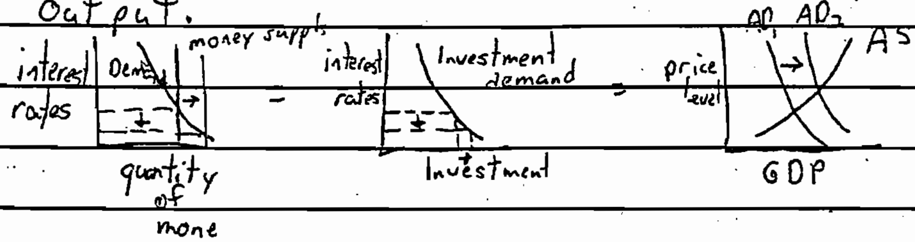
2.) Compitition is reduced domestically. Tariffs increase the price of foreign goods. Therefore domestic goods that may be produced less efficiently than foreign ones (therefore a higher price) will compete with artificially higher priced foreign goods.

b.) I would increase Gov't spending as a fiscal policy and buy gov't securities as a monetary policy.

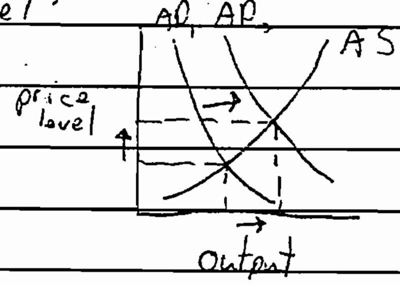
i) ↑ Gov't spending shifts aggregate demand to the right which ↑ output and employment



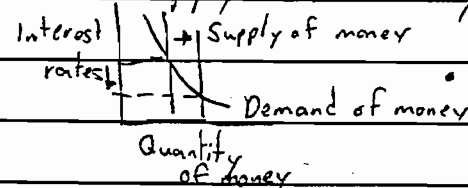
buying securities ↑ the money supply (A) which ↓ interest rates which ↑ Capital Investment which ↑ GDP and Output.



ii) As seen in the previous answer these policies shift Aggregate demand to the right increasing output and price level:

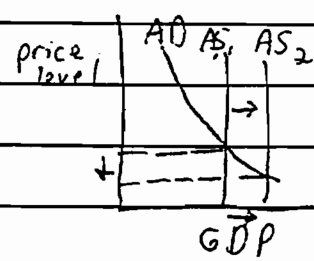


iii) As also seen in section (i), buying securities increases the supply of money ↓ interest rates.



fiscal policy has relatively little effect on the inerest rate although it also helps to ~~incease~~ decreace them.

c.) If decreased interest rates contine in the long run Capital investment spending will increase in the long run as seen in (b, i). Contined Capital spending can lead to increased technology and productivity which shift the long run Aggregate Supply curve to the right ↓ price levels and ↑ output

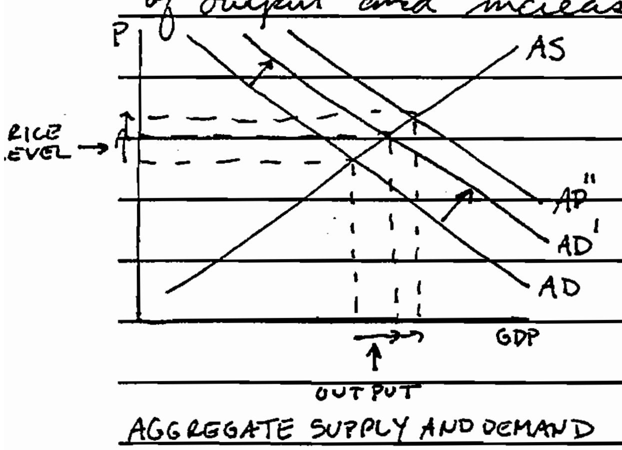


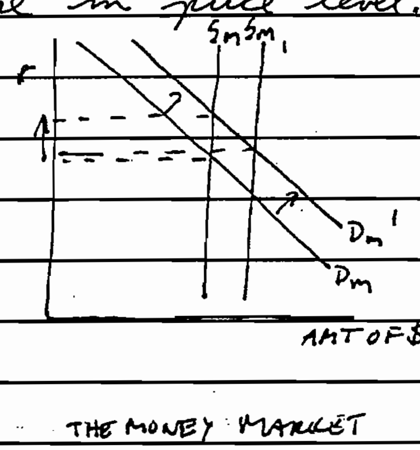
Comment: The candidate correctly identifies two arguments against the use of tariffs and quotas: efficiency is reduced and prices are artificially increased with the reduction in domestic competition. In Part (b), the student correctly describes the effects of expansionary fiscal and monetary policies. However, the interest rate effect of fiscal policy is incorrect. The correct argument is that the increased demand for funds by the government to finance the increase in government spending increases the demand for loanable funds and thus, increases the interest rate. In Part (c), the candidate clearly and concisely presents the argument that the decrease in interest rates (the only interest rate effect identified by this candidate) increases capital investment spending, which can lead to increased productivity and a rightward shift in the long-run aggregate supply curve, which decreases the price level and increases output.

Student Response 3 (Score of 7)

a) ~~Policy I is disadvantageous because even though~~ Even though policy one will solve the problem in the short run, it will fail in the long run. If tariffs and quotas are raised in Country X, every other country around the world will levy tariffs as ~~a~~ ~~[illegible]~~ a response measure. This would cause the net exports to return to their original level and nothing will have been accomplished. Also, by protecting its own industries, Country X is placing them in a less competetive market. Because of this the industries in Country X will further be unable to compete, because the incentive isn't there

b) To reduce unemployment, the government of Country X will engage in expansionary fiscal policy and the Federal Reserve in Country X will practice "loose money." Specifically, the government would increase spending. This additional spending would contribute to a rightward shift in aggregate demand and the associated expansion of output and increase in price level. Because of the





increased aggregate demand, the demand for money to pay for everything will go up. This will result in a raised interest rate. ~~If~~

If the Fed were to buy securities on the open market as part of its "loose money" policy, then aggregate demand will go to the right even more because private investment will go up as a result of the money market's dictation of a lower interest rate. With the shift ~~these two actions solve~~ in aggregate demand, price level and output go up even more, thus solving the problem of unemployment because in order to reach these new levels of output more people must be hired.

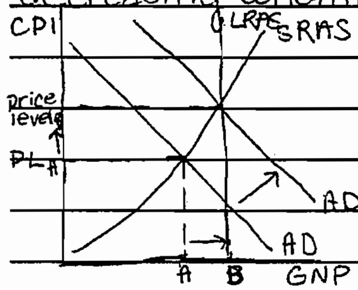
c) The interest rate effects in part (B) nullify each other for the most part. By expanding spending, the interest rate was pushed up, but expanding the supply of money by buying securities lessened that effect or may have even cancelled its effects. This is also evident in the MONEY MARKET graph to the left.

Comment: The student receives full credit in Part (a) for the retaliatory tariffs argument and for reduced incentive to compete in the domestic market after protection. In Part (b) the essay correctly prescribes an expansionary fiscal policy which increases aggregate demand resulting in increases in output and prices. For fiscal policy, the interest rate effect was explained using the increased demand for money. This is a secondary effect which comes from the increase in GDP. The preferred interest rate effect as the increase in demand for loanable funds (or increased supply of Treasury securities) was not discussed. The discussion of monetary policy is fully adequate although not presented in a logical sequence. The increase in money supply leads to a decrease in the interest rate which increases investment. The increase in investment increases aggregate demand which increases output and prices. The answer in Part (c) does not address the question of how the interest rate changes affect economic growth and, hence, receives no credit.

Student Response 4 (Score of 6)

a) One disadvantage of using policy 1 is that the products that now have tariffs (the imports) will be less and the ones that do come into the country will be more expensive. Consumers would be discouraged with this.

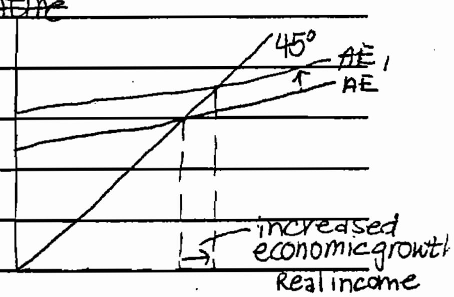
b) To reduce unemployment, the Federal Reserve, using monetary policy would create an easy money situation by purchasing government securities on the open market. This would increase the supply of money, lower interest rates, increase investment and increasing aggregate expenditures; thus, increasing aggregate demand and decreasing unemployment.



(i) Aggregate demand shifts out
(ii) Output increases from A (before shift) to B (after shift) and price level increases from A (before shift) to B (after shift).
(iii) real interest rates ~~increase~~ decrease when the supply of money increases.

A fiscal policy that the government will use to decrease unemployment will be to decrease personal taxes. This will increase disposable income. As shown above, this will increase aggregate demand, price level and output and decrease real interest rates.

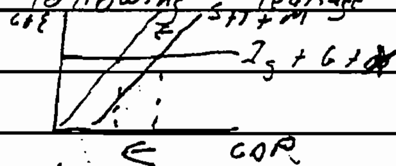
(c) If decreased interest rates would ~~contine~~ continue in the long run, there would be economic growth because there would be increases in investment; thus, increasing aggregate expenditures. The AE curve shifts out with investment expenditures.



Comment: The student receives partial credit in Part (a) for the argument that tariffs and quotas will lead to increased prices in the domestic economy. The discussion of expansionary monetary policy receives full credit in Part (b); however, the discussion of fiscal policy is very brief and there is merely an assertion concerning the interest rate effects of fiscal policy. Thus, only 4 points are earned in Part (b). A point is earned for linking the interest rate decline to an increase in investment to economic growth in Part (c).

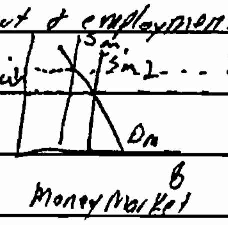
Student Response 5 (Score of 5)

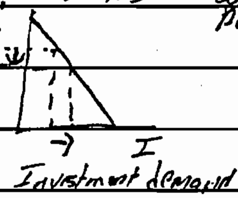
If policy 1. is chosen many problems may occur. One of the most serious problems would be that lower imports will lower GDP. The following leakage-injection model demonstrates this problem.

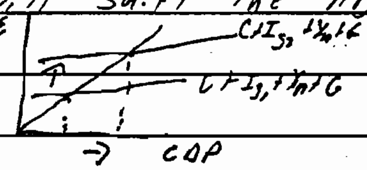


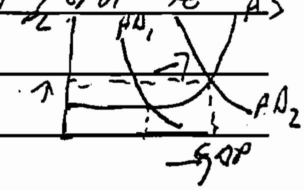
This drop in GDP will further the unemployment problem b/c sales will not be as high and people will lose their jobs.

The most effective monetary policy would be buying bonds in the open market. Buying bonds on the open market will inc. the supply of money. This inc. will cause a dec. in interest rates. This dec. will stimulate Investment which will raise output & employment. This will shift the AD curve to the right.

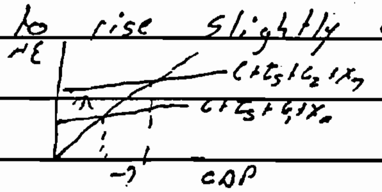


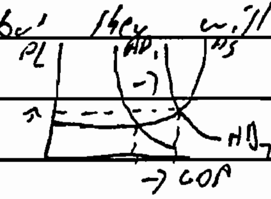


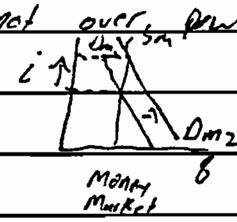




A fiscal policy that will support this action is inc. Gov't Spending. An inc. in gov't spending will cause GDP to rise which will shift the AD curve further to the right. This will cause an inc. in the price level which will create a higher demand for money. This inc. demand will cause interest rates to rise slightly but they will not over power the initial drop.







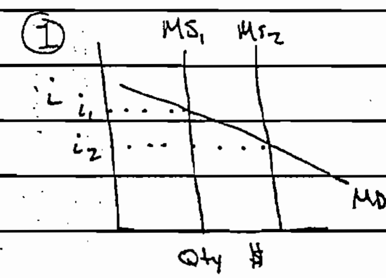
If this policy is continued interest rates will continue to creep back up causing a lower rate of inc. for GDP. The prices will also continue up make the dollar more expensive to other nations which will lower exports and dec. GDP. This will be furthered by an inc. in imports.

Comment: This essay correctly indicates that the decrease in net exports as a result of tariffs and quotas will actually increase the unemployment problem using the leakages-injection model. The candidate receives 1 point for Part (a) because a second disadvantage of tariffs and quotas is not presented. In Part (b), the essay correctly describes expansionary monetary policy and its effect on the economy; the interest rate-investment-aggregate demand linkage is well done. The fiscal policy description is adequate. However, the interest rate effect of fiscal policy is incorrect. Thus, the essay receives only 4 of the 5 points in Part (b). Part (c) is not addressed; the student discusses the international effects of the domestic policy which was not the question on the 1995 exam. Graphical analysis is used to enhance the answer.

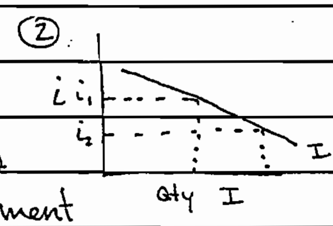
Student Response 6 (Score of 4)

A. The only thing policy 1 does is increase prices of imports and restrict the amounts. the disadvantage is that this is a very limited solution - the effects do nothing to stimulate employment

B. Monetary Policy -



Step one is for FED's to buy securities/bonds this would cause an increase in the banks reserves allowing them more lending power, creating new money therefore increasing the Money Supply (please see graph 1)



as a result of the money supply increasing iii Interest rates will fall (from pt i_1 to i_2 on graph 1) as interest rates fall the quantity of investment increases (see graph 2).

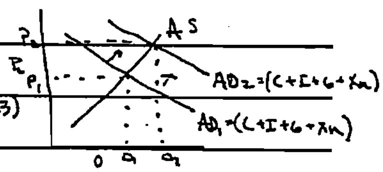
since Investment is a component of Aggregate Demand

i and I increased - AD will increase as a direct result (see graph 3)

the Net results of AD increasing are there will be

ii an increase in Price Levels and an increase in

output and unemployment will therefore decrease.



The fiscal policy that could be taken instead of a monetary policy to reduce unemployment could be

an increase in Government spending

in conjunction with a tax cut

the tax is related to disposable income.

a decrease in Tax results in more disposable

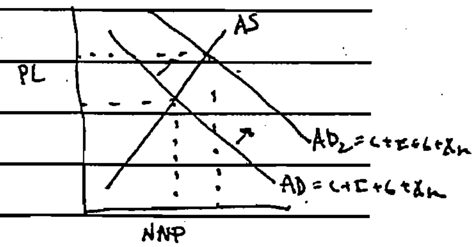
income - allowing greater consumption.

an Increase in consumption + an

Increase in government spending would

result in an increase of Aggregate Demand

(C & G components of AD)



i. Aggregate Demand has increased

ii. Output levels and Price levels have increased

causing Unemployment to decrease

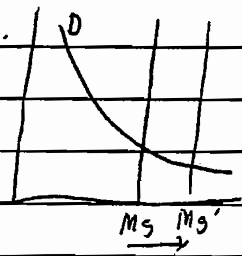
Comment: The answer to Part (a) involves a series of assertions with no explanation as to why tariffs and quotas do not stimulate employment. Expansionary monetary policy is correctly discussed with its effects via the interest rate and investment. The graphical analysis amplifies the essay. The fiscal policy description is good. However, the interest rate effect of fiscal policy is not addressed. Part (c) is not attempted. Thus, the student receives no credit for Parts (a) and (c), and receives 4 of the 5 points in Part (b).

Student Response 7 (Score of 3)

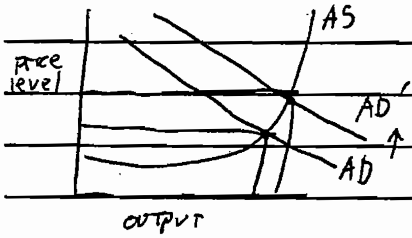
a) Restricting imports may protect jobs in country X. However, if Country X does not have appropriate natural resources, does not have enough of these resources, or is technologically behind in production of these resources, then domestic production could decline for these products. The tariffs and quotas interfere with the positive aspects of comparitive advantage, and output decreases.

In addition, Country X's economy may suffer if exports decline as a result of trade barriers in other countries.

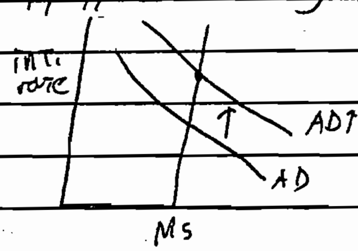
b) The best monetary policy action would be the government's buying of securities. In the money market, This increases ~~gov't~~ demand for dollars, ~~and increases the real interest rate:~~ Thereby increasing money supply and decreasing the interest rate.



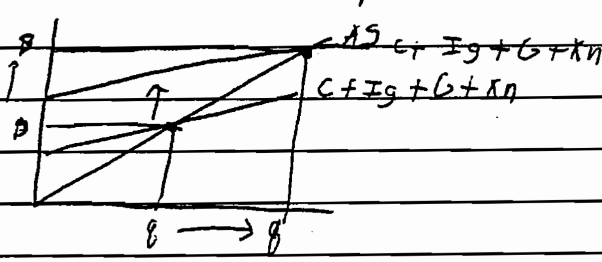
Increased money supply is in the form of increased excess reserves. Banks can therefore loan out more money and expand the economy. Output and price level will increase, if in the intermediate range of AS.



An example of fiscal policy would be increased government spending. This increased demand on a steady money supply, increasing interest rates.



Investment spending increases as a result of higher interest rates. Output increases; as does spending.



Price level increases as well.

c) Lower interest rates as a result of monetary policy will result in loss of foreign investment and the depreciation of the dollar. Economic growth will decline, ~~also as a result of lower~~

Higher interest rates as a result of ~~mone~~ fiscal policy will cause an increase in foreign investment and appreciation. However, net exports would eventually decline, slowing economic growth relatively.

Comment: The essay is inadequate. The student receives credit for the possibility of retaliatory trade barriers. In Part (b), the student identifies expansionary monetary and fiscal policies as the correct stabilization policy prescriptions which earns the student 1 point. An additional point is awarded for the effect on aggregate demand of expansionary monetary policy, although the student misses the interest rate-investment relationship. In Part (c), the student confuses foreign investment with capital investment and thus receives no credit.

Student Response 8 (Score of 2)

As the rate of unemployment all most doubles from 5 percent to 9 percent I have been given to options:

policy 1: Use tariffs and quotas to restrict imports and thus protect jobs in Country X

policy 2: use monetary and fiscal policies to solve the unemployment problem without resorting to trade restrictions.

To become isolated and ~~put~~ ~~tariffs~~ ~~and~~ increase tariffs and put quotas would be a disorder because we would put added pressure on companies and people to build and produce other products that we now limited

It would all so discourage trade with other countries. If we increase tariffs other countries would all so react in the same way. Thus giving us a overabundance of goods.

The way to solve this problem is thru a monetary policy or fesical policy action. First you have to stimulate the economy by increasing government spending and decreasing taxes, this is a fiscal policy which would, incourage companies to reinvest and expand. You would increase the amount of money out there by having the Federal Reserve drop the requirement which gives banks more money to lend out and more jobs can be created. A increase in the discount rate for a long time would be a foolish thing to do because then everyone would want to invest instead of expanding and making companies more efient.

Comment: At this end of the scale, students indicate a great lack of economic knowledge. This essay earns 1 point in Part (a) for mentioning the possibility of retaliatory trade policies and 1 point for correctly identifying an expansionary monetary policy and expansionary fiscal policy.

Student Response 9 (Score of 1)

a) Policy 1: Use tariffs and quotas to restrict imports and thus protect jobs in Country X.

This solution makes perfect sense but does not look at the other possible outcomes of such tariffs and quotas. The jobs within country X may be protected, but they will also be limited to what is available within country X. Bringing in more imports could also bring in more jobs. Protection of jobs is not going to help anybody if there is such a limited number of jobs to be protected due to lack of imports. All imports must be handled in some way. Increased imports means increased manpower needed to handle the incoming imports. Tariffs and quotas will only restrict the potential employment growth with decreased imports.

Another disadvantage to using tariffs and quotas to limit imports is the possible negative reaction from the other countries wishing to export to country X. Other countries wishing to trade may not want to wait around for country X to develop a stable rate of unemployment. Businesses may be taken to other countries that are willing to trade at that same time.

b) A monetary policy that could possibly combat the unemployment problem, would be to lower the income tax rate. By doing this, the job becomes more valuable to the employee and to the unemployed individual seeking work. People would have a greater desire to work knowing that they will be able to retain a substantial amount of what is earned. (i) the demand of jobs would go up due to the increased interest in employment (ii) output and price levels will remain relatively constant due to the little change in supply/demand for the short term (iii) interest rates would slightly increase due to the increased amount of currency possesed by the individuals possibly looking for investment.

A fiscal policy would possibly be to lower government spending on things other than employment research. The government should put more money toward employment projects/agencies to help promote employment within the state communities. (i) With increased government spending going toward employment projects/agencies the demand for employment by both parties will increase (ii) output and price levels will remain relatively the same until employment rates dramaticaly increase (iii) interest rates will go up due to the spending of government money that must be compensated for.

c) If higher interest rates continue into the long run, the economy may remain the same until employment increases. But while the interest rates increase, the government will continue to have this income of money and therefore will be able to help the country as a whole.

Comment: This candidate's response is extremely inadequate. The response consists of nearly two pages of incorrect statements, as in the explanation of monetary and fiscal policy, or statements unrelated to the question, as in the first paragraph on tariffs and quotas. The candidate does address, in a limited way, the possibility of a "negative reaction"— trade war/retaliation and for this receives a point. This is an excellent example of an application of the holistic approach to scoring; the reader looks for a correct relevant statement.

Free-Response Question 2

Assume that the economy is in a recession.

(a) Explain each of the following.
  (i) Monetary and fiscal policies advocated by monetarists to eliminate the recession
  (ii) Monetary and fiscal policies advocated by Keynesians to eliminate the recession

(b) Explain how monetarists and Keynesians differ in their conclusions about the effects of crowding out associated with the stabilization policies outlined in Part (a).

Question 2 Scoring Guide

Basically the point distribution is 3 points for Part (a); 2 for Part (b).

Part (a): 3 points

Monetarists would advocate a money rule with no discretionary monetary or fiscal policy. They believe that fiscal policy is neutral due to crowding out. The monetary rule is growth of the money supply at the growth rate of real GDP, at 3-5% a year, or at an announced constant rate.

Keynesians believe that an expansionary fiscal policy should be used to eliminate a recession. Keynesians, in general, think that monetary policy is "unreliable" or less effective than fiscal policies. Low interest rates will not necessarily induce increased business investment or consumer spending.

Points:

1 — Student demonstrates an understanding of the monetary rule. Student must do more than state that the monetarists have a monetary rule, but does not have to explain it in detail.

½ — Monetarists believe fiscal policy is ineffective.

1 — Keynesians advocate an expansionary fiscal policy to eliminate a recession.

½ — Fiscal policy is more effective than monetary policy, or monetary policy is ineffective, or monetary policy is complementary to fiscal policy.

Part (b): 2 points

Monetarists believe that there is complete crowding out, thus making fiscal policy ineffective. Keynesians believe that there is little or no crowding out. Crowding out occurs when the government borrows money to increase government spending and thus reduces the amount of loanable funds available to the private sector.

Alternatively, the government enters the loanable funds market and is willing to pay whatever interest rate is necessary to "call forth" the funds required. The increase in interest rates reduces investment in the private sector.

Some students may discuss the differences between Keynesians and monetarists in terms of the interest sensitivity of investment — this is acceptable if correct.

Points:

1 — Statement that monetarists believe that there is crowding out and Keynesians believe there is little or no crowding out.

1 — Explanation of why monetarists and Keynesians take these positions. This explanation needs to demonstrate an understanding of crowding out.

Besides counting points, the answer may be looked at as a whole and ultimately judged by its overall quality. This is particularly true if the total score includes a half point. The final total should mean something in terms of the overall quality of the answer. 5 should reflect an excellent answer but not necessarily perfect; 4, an excellent answer with a flaw; 3, a good answer; 2, an adequate answer; 1, a seriously deficient answer, but still an answer; 0, all else.

Overall Comment on Student Responses to Question 2

The first short answer question asked the student to focus on the difference between the monetarist and Keynesian policy prescriptions for an economy in a recession. This question emphasized the need for knowing two important schools of economic thought and, in particular, the stabilization policies advocated by the different schools of thought. Many students responded by providing a discussion of expansionary stabilization policies, completely ignoring the viewpoints of Keynes and the monetarists. There was also evidence that students are unable to explain the crowding-out effect; although they may know the monetarist and Keynesian views of crowding out.

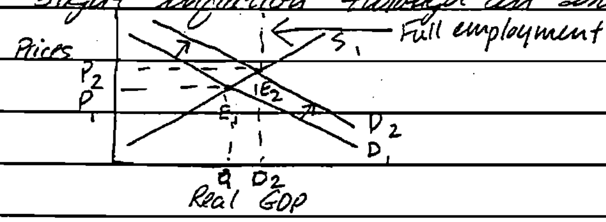
Sample Student Responses

Student Response 1 (Score of 5)

(a)

(i.) Monetarists would advocate very ~~[illegible]~~ limited monetary and fiscal policies during a recession. In this way they are quite similiar to Classical economists. They would ~~want to~~ have a monetary policy which simply increased the money supply 3-5% a year. No other monetary action would be taken. The monetarists would propose little in the way of fiscal policy as well. Since monetarists believe that fiscal policy has a negligable effect on the economy in the long run.

~~[illegible]~~ (ii) Keynesians believed strongly in the use of fiscal policy, and to a lesser extent monetary policy. For monetary Policy, Keynesians would advocate either buying gov. securities, lowering the discount rate, or lowering the reserve requirement. The latter two policies have very powerful effects, however. Most often, Keynesian monetary policy relies on open-market operations. Keynesians would propose strong fiscal policy. They would combine a cut in taxes with an increase in government spending. This would increase employment and overall output; but probably also lead to slight inflation through an increase in demand.



(b) Monetarists believe that the crowding out effect virtually eliminates any effects of activist fiscal policy. The increase in gov. spending would simply drive interest rates up and force consumers out of the market. Keynesians do not feel that crowding-out is as serious of a dilemma. Though they acknowledge that crowding out does occur, they do not believe that ~~a~~ total crowding out will result. For Keynesians, an increase in gov. spending leads to an increase in consumer demand through ~~the~~ interest rates

Comment: This essay clearly indicates that monetarists would not use discretionary policy; monetarists would follow a monetary rule of a 3-5% per year increase in the money supply; and monetarists believe that fiscal policy is ineffective in the long run. The essay states the Keynesian position well with strong fiscal policy and the use of monetary policy as ancillary to the fiscal policy. In Part (b), the essay states the monetarist view of total crowding out and the Keynesian view that the crowding out would be incomplete. In addition, the student demonstrates an understanding of crowding out by describing the increase in government spending leading to a rise in interest rates which, in turn, drives consumers out of the market.

Student Response 2 (Score of 4)

2a) i) Monetarists would focus more on the monetary policies to counteract the recession and expect fiscal policies to reinforce what they do. Monetarists would recommend the purchase of govt. securities to increase the money supply. They would also recommend a drop in the prime rate and the reserve requirement as well to increase the money supply. Fiscal policy would be that of no policy, keeping Govt. expenditures the same as well as taxes. This monetary policy is intended to emphasize investment spending to increase AS and hire more people.

ii) Keynesians would on the other hand advocate fiscal policies to solve the ~~as~~ recession with monetary policies to reinforce that. Since Keynesians had no fear of inflation they would recommend a means of increasing AD. They would do this by increasing govt. expenditures and cutting taxes, effectively running a budget deficit. This would allow both G and consumption portions of AD to increase resulting in more output, and the hiring of more workers.

2b) Monetarists would claim that Keynesian tactics would crowd out investment. They state that since government is using up all the money, private firms will have to pay high interest rates to get the money needed for investment. Keynesians are highly suspicious of the true nature of the "crowding out effect" and claim it is negligible. Monetarists would claim that the "crowding out effect" will have the money inefficiently used by the govt. instead of effectively used by the private sector

Comment: This essay loses a point in Part (a) by not correctly stating the monetarist position. The essay indicates that monetarists would use discretionary monetary policy; this is incorrect. However, the student continues and indicates, correctly, that monetarists would advocate using no discretionary fiscal policy. The essay on the Keynesian policy prescriptive correctly identifies discretionary fiscal policy with monetary policy as a reinforcement. In Part (b), the monetarist position of discretionary fiscal policy crowding out private investment is clearly stated with the Keynesian perspective of a negligible impact of crowding out. Further, the essay indicates an understanding of crowding out.

Student Response 3 (Score of 3)

2. Economy in recession
a) Effects of monetary + fiscal policies of monetarists v. Keynesians
1. monetarist policies
a. decrease interest rates
1. increases investment
b. expand supply of money
1. more available money increases demand and GNP
c. lower reserve requirement
1. makes more money available ~~to~~ for banks to len
d. purchase bonds on open market
1. makes money more available
2. Keynesian policies
a. increase federal spending
1. more federal spending equates to more jobs and more money injected into economy
2. decrease in taxes
1. public has more money to spend in economy
b) Differing views between two groups regarding effects of crowding out

1. ~~Keynesians believe that investment~~ Monetarists believe increases in federal spending, resulting in higher interest rates, crowds out potential investors who would help the economy in the long run.

2. Keynesians believe that there is no substantial crowding out of investors

Comment: The student is denied 1 point in Part (a) for incorrectly discussing expansionary monetary policy rather than the monetary rule of increasing the money supply at the rate of 3-5% a year or an announced constant rate. The student was denied a ½ point for not stating the monetarist position that fiscal policy is ineffective. The student was granted 1 point for correctly asserting that the Keynesians would use fiscal policy, increasing federal spending, and decreasing taxes. The student was denied a ½ point for not stating that Keynesians believe that fiscal policy is more effective than monetary policy or that monetary policy is ineffective or monetary policy should be complementary to fiscal policy. In Part (b), the student was granted full credit for two reasons: firstly, for asserting that monetarists believe that increases in federal spending crowd out potential investors and Keynesians believe that there is no substantial crowding out of investors, and secondly, for demonstrating an understanding of crowding out by describing crowding out.

Student Response 4 (Score of 2)

2a. i. Monetarists believe that active fiscal policies seriously destabilize the economy, worsening the swing of the business cycle. Therefore, they would advocate little to no fiscal action.

Depending on who you're talking to, monetarist want monetary policy either to be slightly expansionary in a recession ~~selling~~ buying bonds and jawboning mostly, or to simply let the market system run its course, conforming to a simple and constant monetary growth rule, increasing the money supply by perhaps 4% a year.

ii. Keynesians would advocate an all out expansionary fiscal & monetary policy, increasing government spending, perhaps decreasing taxes, and attempt to expand the money supply. However, the monetary policy would be considered a distant second measure, because via the ~~[illegible]~~ multiplier, fiscal policy is ~~much~~ more effective.

b. Keynesians do not account for crowding out, either they don't believe in it, just as monetarist don't believe in liquidity traps, or they ignore it. Monetarists feel that as government spending increases, it "crowds out" private sector production, effectively nullifying a significant portion of the Keynesian's action, suggesting the greater importance of monetary policy.

Comment: The opening paragraph of the essay states correctly that monetarists would not advocate government intervention (fiscal policy) to solve a recession. However, the student continues with an explanation that monetarists would advocate an expansionary monetary policy to stimulate the stagnating economy. This is incorrect. Thus, the student earns a 1/2 point for the discussion of the monetarist position. The discussion of the Keynesian position of expansionary fiscal policy along with discretionary monetary policy earns full credit (1 1/2 points). The remaining portion of the essay incorrectly addresses crowding out and thus, receives no credit.

Student Response 5 (Score of 1)

(a) In a recession, policy makers seek to stimulate the economy by employing easy-money, or expansionary, methods. The monetarists advocate the use of monetary policies only to influence the economy and try to influence aggregate demand. By influencing demand, they believe, the aggregate supply is also changed. In a recession, monetarists would advocate using open market operations. The Federal Reserve would buy bonds to inject money into the market. Or, the reserve requirement of commercial banks would be reduced, thus increasing the amount of excess reserves as in the equation ~~[illegible]~~ below:

TR − ↓RR = ↑ER which leads to ↑ money supply

Banks would have more funds available to lend. Also, monetarists would encourage the lowering of the discount rate - the interest the Fed charges on commercial banks' loans. This, too, would increase the available amount of money.

(b) Fiscal and monetary policies, or a combination of both, would be used by the Keynesians to influence both AS + AD. Along with the methods above, they would use government spending increases. This injects money into the economy and encourages increased output and employment. Also, reducing taxes would have the effect of stimulation because consumers would have more money for expenditures. These stabilization policies are all expansionary measures to reduce the effects of a recession.

(b) Keynesians believe that

Comment: This essay indicates knowledge of expansionary discretionary stabilization policies but the writer does not know the monetarist position sufficiently to receive any credit. The student does receive credit for the explanation of Keynesian policy prescriptions, however crowding out was not addressed. The student earned a single point because the answer in its entirety does not respond to the question.

Free-Response Question 3

Explain how some individuals are helped and others harmed by unanticipated inflation as they participate in each of the following markets.

(a) Credit markets
(b) Labor markets
(c) Product markets

Question 3 Scoring Guide

Basically the point distribution is 2 points for Part (a), 2 for Part (b), and 1 for Part (c).

Part (a): 2 points total

With unanticipated inflation, borrowers are helped in that they repay in dollars which have less purchasing power than the dollars they borrowed. Likewise creditors are hurt because they receive payment in dollars which have less purchasing power. The student must assert who is helped <u>and</u> who is hurt.

Points:
1 —For assertion.
1 —For explanation.

Part (b): 2 points total

Employers benefit if output prices rise faster than labor costs. The unanticipated inflation hurts employees if their real wages have decreased. (Possible to receive a $^{1}/_{2}$ point.)

Points:
1 —Employer's impact with explanation.
1 —Employee's impact with explanation.

Part (c): 1 point total

If producers can raise prices faster than costs, they are helped. If producers have long-term contracts on prices, then they are hurt because costs could rise faster than prices. Consumers are hurt because real income is falling.

1 — Correct argument with explanation.

Besides counting points, the answer may be looked at as a whole and ultimately judged by its overall quality. This is particularly true if the total point count includes a half point. The final total should mean something in terms of the overall quality of the answer. 5 should reflect an excellent answer but not necessarily perfect; 4, an excellent answer with a flaw; 3, a good answer; 2, an adequate answer; 1, a seriously deficient answer, but still an answer; 0, all else.

Overall Comment on
Student Responses to Question 3

The second short question on the AP Macroeconomics Exam focused on the effects of unanticipated inflation on credit, labor, and product markets. The students did not appear to know who the players were in each of the markets, and the language of "markets" appeared to confuse the students. The students did seem to have a firm understanding of the impact of unanticipated inflation on the credit market.

Sample Student Responses

Student Response 1 (Score of 5)

In a credit market, creditors are hurt by unexpected inflation, and debtors are helped. Inflation causes a decrease in the value of the dollar, which means the debtors are paying their debts back in dollars worth less than those they borrowed.

In a labor market, laborers are hurt by inflation and employers are helped. The inflation causes the wages of the laborers to be worth less and have less buying power, while the employer is paying the same wages to the workers, in dollars that are suddenly worth less.

In a product market, buyers are hurt by inflation because they can suddenly buy less with their dollars. The producers can be hurt by this as well. While they are now getting a higher price for their products, they are selling fewer because consumers cannot afford as many.

Comment: This is an excellent essay written without using economic jargon. The student covers the credit and labor markets very well. The student earns the point for Part (c) for concluding that consumers (buyers) are hurt because their real income is falling. The second statement in Part (c) is not relevant to the question but no points are deducted. Only one correct argument is required for Part (c). Note: this answer indicates that a 5 is not a perfect answer; rather it is an excellent answer.

Student Response 2 (Score of 4)

(a) Credit- loaners are hurt because when debtors payback, they are actually paying back a value of money worth less than that they borrowed.

For example- If somebody borrows $100 from a bank and must pay it back in a year where there is 1% inflation, he is only really paying back $93 (or 93%) this hurts the creditor and helps the borrower.

(b) Labor - The employer benefits and the employee is hurt because the real wage that the employee receives is worth less than its stated amount relative to the price level increase. Employer is actually paying lower value wages while the price level and supply increases.
(c) Product - The general price level in the product market increases due to inflation. Consumers actually increase and speed up the level of inflation by trying to avoid inflation by buying large quantities of the product. (Demand-Pull inflation)

Comment: In Part (a), the student correctly identifies that debtors are helped by unanticipated inflation in that they pay back loans in dollars with lower purchasing power and that creditors are hurt by this. The answer receives 2 points for Part (a). In Part (b), the impacts in the labor market are correctly identified. The answer receives 2 points in Part (b). In Part (c), the student earns no credit since the answer of demand-pull inflation is incorrect.

Student Response 3 (Score of 3)

(a) Creditors would lose because the purchasing power of the money they lent will be lower than when the lent it.
(b) Labor markets will be hurt because real wages will be lower and unless they have a COLA agreement they will not rise.

(c) Product markets will be harmed because goods are more expensive and people real income has fallen

Comment: The candidate correctly identifies that creditors will be hurt with unanticipated inflation because of the reduced purchasing power of the loan payments. However, the essay does not address the effect on the debtors; thus, the essay earns 1 point for Part (a). The labor market effect of lower real wages is correctly identified. Once again the impact on employers is not addressed in the answer and, thus, the essay earns 1 point in Part (b). The real income argument is presented and earns 1 point in Part (c).

Student Response 4 (Score of 2)

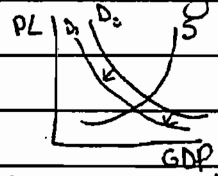
In the credit market only the borrowers are benefited from unanticipated inflation. This is because the value of the money they have borrowed is lessened and not as much on them to pay back. Consequently this hurts the lenders because they are in effect not recieving the total value of what they lent out.

The labor market is hurt by inflation because the prices of their products rise in accord with the rise in cost of making them that inflation induces. This lowers their aggregate demand because coupled with the inflation consumers can't afford their product and consumption drops.

Comment: The student correctly identifies the effects of unanticipated inflation in the credit market in Part (a), and thus receives the 2 points. In Part (b), the student does not correctly identify the effects, and Part (c) is not answered. This essay represents a very typical answer to the third macroeconomics question.

Student Response 5 (Score of 1)

Unanticipated inflation ~~may both~~ will help ~~and~~ or harm different groups. Credit markets will be hurt by unanticipated inflation because they ~~be~~ will receive in loan payments cheaper dollars than they loaned. Labor markets will be hurt by unanticipated inflation. As the price level rises, the demand for some goods and services will decrease. As the



demand decreases, the quantity demanded will decrease, as with the quantity supplied. Businesses will not need as many workers to produce their products. Product markets will at first be helped by unanticipated inflation because they will be able to sell their goods and services at higher prices. Eventually, the product markets might be hurt because as the level of unemployment rises, the total GDP declines.

Comment: In this essay, the student correctly identifies that creditors will be harmed by unanticipated inflation and earns 1 point for this statement. The student says "credit markets," but the context of the answer makes it clear that he or she meant "creditors." However, the essay does not correctly identify the other ways in which unanticipated inflation affects the credit, labor, and product markets.